



Getting Started with Google MCP Servers

Google AI Community, December 17, 2025



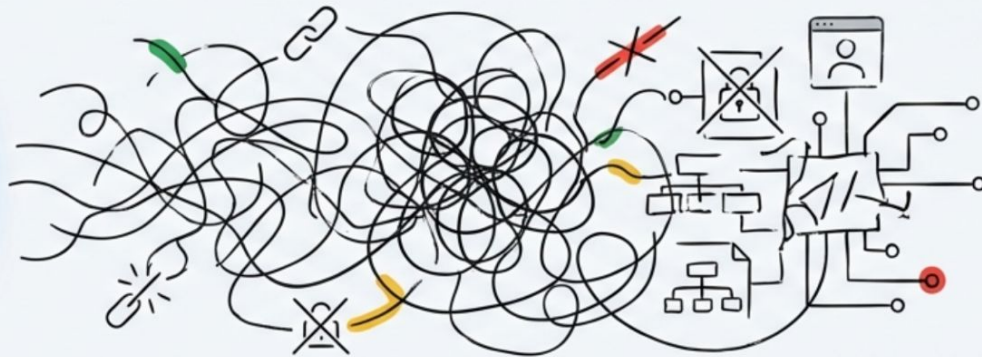
Powerful AI models are brilliant, but isolated.

Connecting large language models to real-world data and services is the biggest challenge in building useful AI agents. **Traditional methods rely on brittle API wrappers, complex authentication flows, and manual schema definitions.**

We call this the 'last mile problem'.



AI Core



Real-World Services

The industry is standardising on a universal language for AI.

The Model Context Protocol (MCP) is the open standard emerging as the industry-sanctioned solution. It standardises the interface between AI models and external systems, ensuring agentic systems are grounded in real-world data.



AI Core



**Model Context
Protocol (MCP)**



Real-World Services

Google now offers fully managed MCP services for its core cloud products.

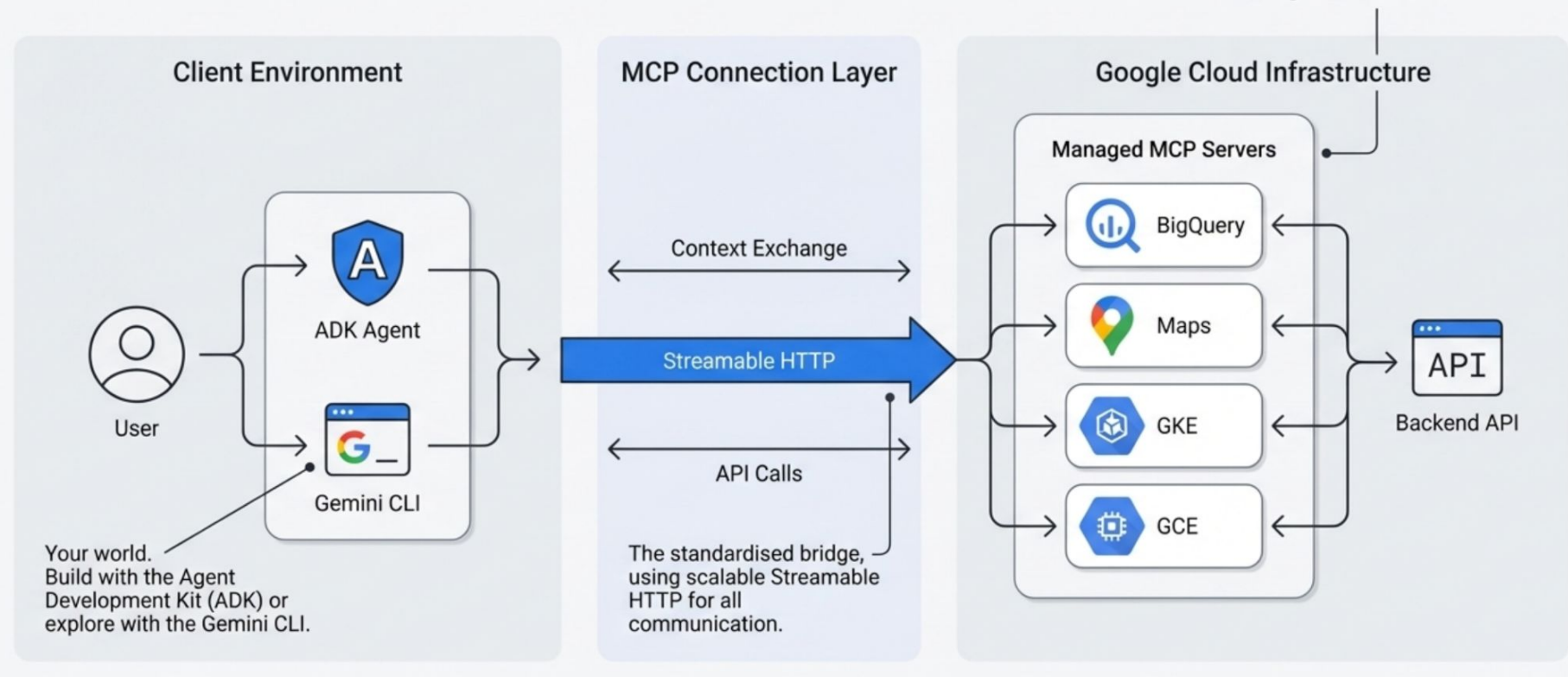
Announced on December 10, 2025, this changes everything. Instead of writing wrappers or hosting servers, developers can now point AI agents to globally-consistent, enterprise-ready endpoints for Google's most powerful services.



Google
Maps Platform



Here's how your agent connects to Google's infrastructure.



Your agent now speaks BigQuery, Maps, GKE, and GCE.



BigQuery

The Analytical Engine. Turn natural language questions into SQL execution on petabyte-scale data.



Google Maps

Geospatial Intelligence. Ground your agent in the physical world to prevent location-based hallucinations.



GKE

The DevOps Operator. Give your agent read-access to your Kubernetes control plane for intelligent monitoring.



GCE

Infrastructure Management. Allow your agent to query and interact with virtual machine primitives.

This is more than an API endpoint. It's an enterprise-grade solution.



Scalable Transport

Uses Streamable HTTP for stateless, scalable interactions compatible with standard load balancers and security proxies.



Simplified Networking

Each service has a uniform endpoint (e.g., ``bigquery.googleapis.com/mcp``), simplifying network configuration and firewall rules.



Native IAM Security

Integrates directly with Google Cloud IAM. No more passing API keys in prompts. This is true enterprise-grade security.

Enabling the BigQuery MCP Server

Service Enablement & Dual-Layer Security Setup

⚙️ . Enable Required APIs

You must enable both the underlying BigQuery API and the specific MCP interface

```
# Enable the standard BigQuery Service
gcloud services enable bigquery.googleapis.com
```

```
# Enable the MCP Interface (Beta)
gcloud beta services mcp enable bigquery.googleapis.com
```

🛡️ 2. Security Gate 1: Grant MCP Access This grants the identity permission to "talk" to the MCP server itself.

```
gcloud projects add-iam-policy-binding $PROJECT_ID \
  --member="user:$USER_EMAIL" \
  --role="roles/mcp.toolUser"
```

🛡️ 3. Security Gate 2: Grant Service Access This grants the identity permission to view the actual data within BigQuery.

```
gcloud projects add-iam-policy-binding $PROJECT_ID \
  --member="user:$USER_EMAIL" \
  --role="roles/bigquery.dataViewer"
```

Your agent must pass two gates to execute any action.



Using MCP Inspector for testing

1. Ensure that APIs and MCP Servers are enabled in the project. Local Authentication is done too.

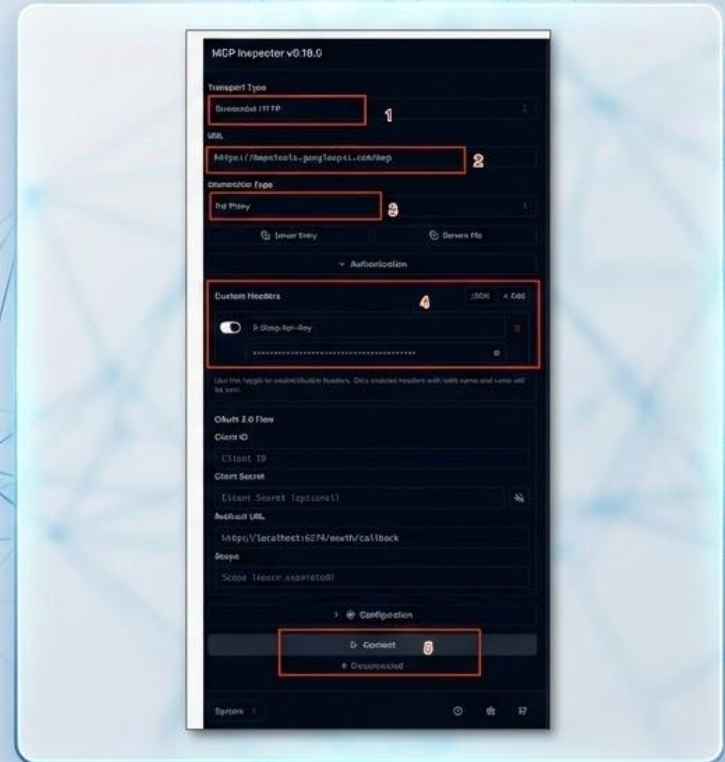
2. Launch MCP Inspector

```
(npx @modelcontextprotocol/inspector)
```

3. Connect to MCP Server

4. List tools

5. Invoke tool





Configuring Google MCP Servers in Gemini CLI

Create an MCP extension for Gemini CLI : `~/.gemini/extensions/bqmcg/gemini-extension.json`

```
{
  "name": "bqmcg",
  "version": "1.0.0",
  "mcpServers": {
    "bigquery-mcp-server": {
      "httpUrl": "https://bigquery.googleapis.com/mcp",
      "authProviderType": "google_credentials",
      "oauth": {
        "scopes": ["https://www.googleapis.com/auth/cloud-platform"]
      },
      "headers": {
        "x-goog-user-project": "YOUR_PROJECT_ID"
      }
    }
  }
}
```



From natural language to intelligent action, seamlessly.



“I am planning a drive from Mumbai to Pune tomorrow. What should I expect?”

```
> Agent thought: The user wants information about a drive.
I should check the weather and compute the route.

> Calling tool: lookup_weather (maps-grounding-lite-mcp)
{
  "location": "Mumbai to Pune",
  "date": "tomorrow"
}

> Calling tool: compute_routes (maps-grounding-lite-mcp)
{
  "origin": "Mumbai",
  "destination": "Pune",
  "departureTime": "tomorrow"
}

> 
```

The agent combines multiple tools to give a complete answer.



The Drive

- Distance: Approximately 154 km.
- Duration: Expect the drive to take around 3 hours.



The Weather

- Mumbai: Warm and pleasant. High: $\sim 30^{\circ}\text{C}$.
- Pune: Sunny but much cooler. High: $\sim 28^{\circ}\text{C}$.

This entire interaction was powered by the Google Maps MCP Server. No custom code required.

Your agent can now be your data analyst.



Find the release notes from
'bigquery-public-
data.google_cloud_release_notes'
for 'Google Kubernetes Engine'

Product: Google Kubernetes Engine

Date	Description
Date: 2024-07-15	Introduced GKE Autopilot support for larger clusters. General availability.
Date: 2024-06-28	Security patch release. Fixed a vulnerability in container runtime.

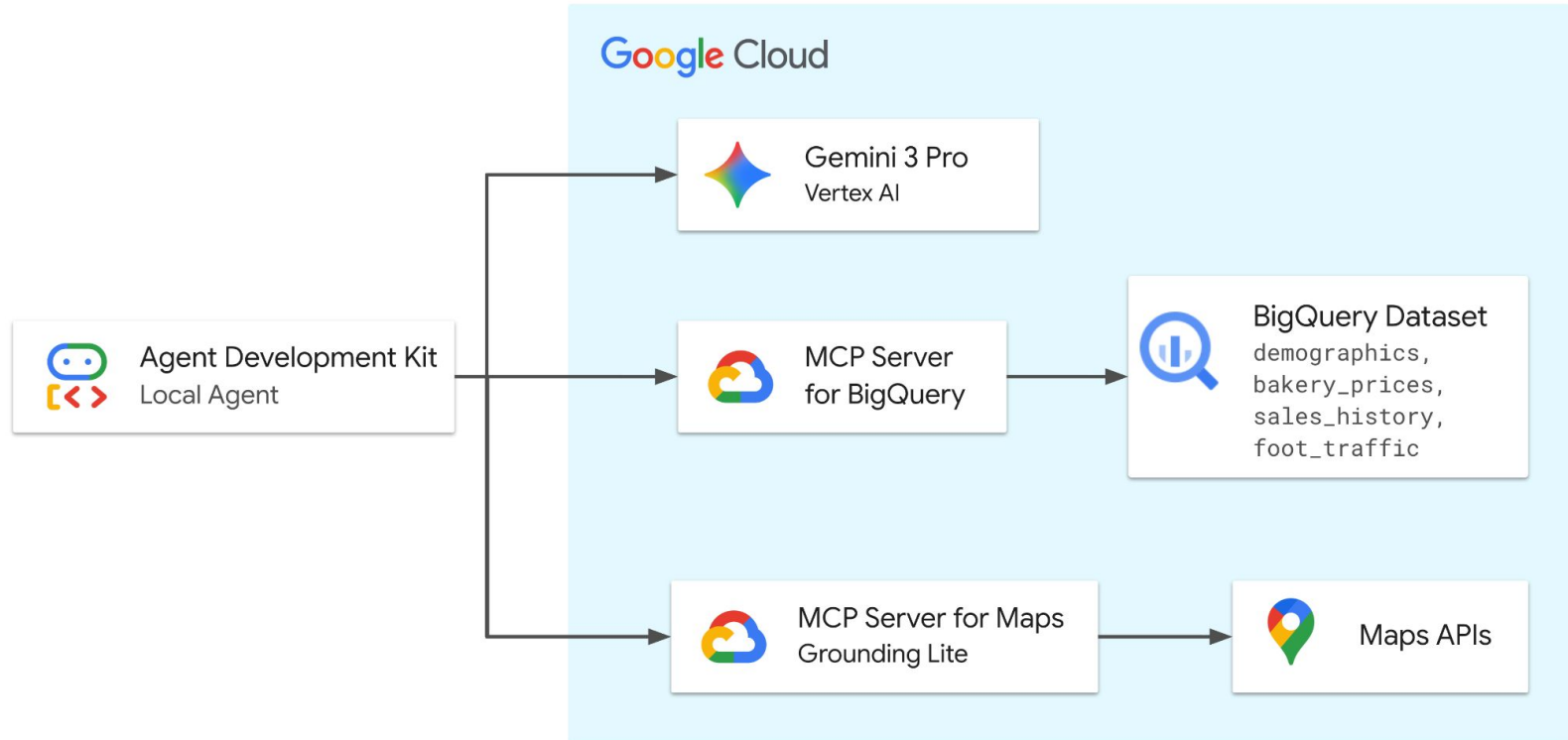
The agent used the ``execute_sql`` tool on the BigQuery MCP Server to translate the prompt, run the query, and format the result.

Ready to build your own? The Agent Development Kit is your starting point.

The Google Agent Development Kit (ADK) lets you build custom agents in Python that integrate seamlessly with MCP Servers. Define custom instructions and chain tools to solve complex business problems.

```
# Initialize the Toolset for a remote MCP Server
maps_toolset = McpToolset(
    connection_params=StreamableHTTPConnectionParams(
        url=MAPS_MCP_URL,
        headers={"X-Goog-API-Key": MAPS_API_KEY}
    )
)
```

Agent Demo (Cross Product Queries)



Sample queries



Market Research & Location

Find the zip code with the highest morning foot traffic score.
Search for 'Bakeries' in that zip code to see if it's saturated.
Check for 'Specialty Coffee' shops to position near them.



Financials & Forecasting

What is the maximum price for a 'Sourdough Loaf' in the LA Metro area?
Run a forecast for December 2025 sales quantity based on history.
Calculate projected total revenue using a premium price (\$18).



Logistics & Operations

Find the closest 'Restaurant Depot' to the area.
Verify drive time is under 30 minutes for daily restocking.



The era of the secure, governed, and capable AI agent is here.



1. Enable the Endpoints: Activate MCP for BigQuery, Maps, GKE, and GCE in your project.



2. Configure Identity: Use IAM for secure, keyless access control.



3. Adopt the Frameworks: Start with the Gemini CLI for exploration, then graduate to the ADK to build custom agents.

Dive deeper with our advanced codelab: **Build a Location Intelligence Agent with MCP servers for BigQuery and Google Maps.**



